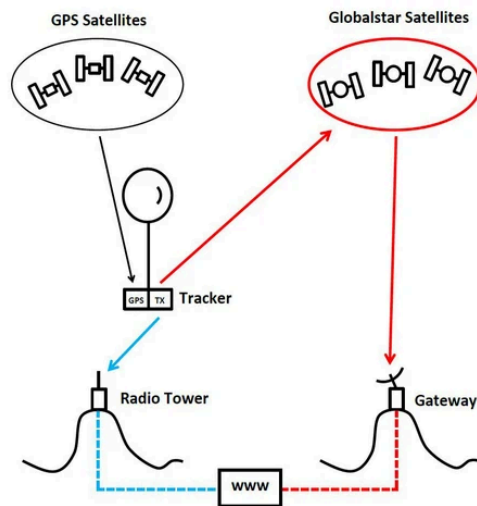


Tracking a Weather Balloon

When it comes to tracking a weather balloon, the three most common options are a satellite tracker, an APRS tracker, or a cell phone. Each option has its benefits and limitations. A cellphone tracker is illegal and should never be used.

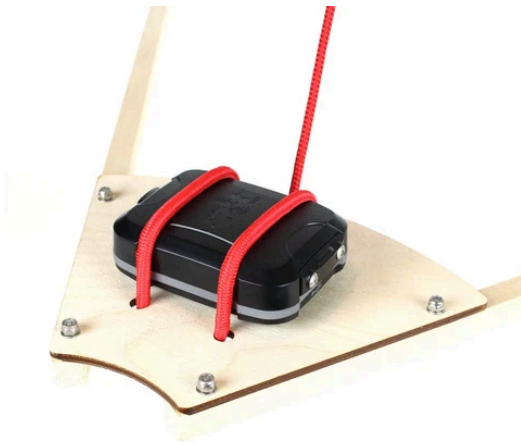


Basic Understanding of GPS technology - One of the key technologies that have allowed us to track and recover weather balloon payloads is the Global Positioning System or GPS for short. GPS was developed by the U.S. Department of Defense for military purposes. It was never originally intended for civilian use. In 1983 an airliner carrying 246 passengers (**Korean Airlines Flight 007**) was shot down by Russia after accidentally straying into their prohibited airspace. This prompted President Ronald Reagan to issue a directive making GPS freely available for civilian use to prevent future navigational errors that resulted in loss of life. However, it is important to understand that civilians do not have access to all of the features of the GPS system. Most civilian GPS receivers will stop working at an altitude of roughly 18,000 m ~ 60,000 ft. This prevents most trackers from being able to update your payload's position above 18,000 m. Sometimes it is possible to obtain specialized GPS receivers (more expensive) that work above 18,000 m. The GPS receiver on our **Eagle Flight Computer** (square block in center of flight computer) is a specialized GPS receiver that works at extremely high altitudes.



How a Tracking System Works. In order to have a tracking system, you need a tracker, a network, and an internet connection. A tracker is simply a GPS receiver and a radio transmitter built into a single piece of hardware. The tracker is able to determine its precise location by receiving position signals from GPS satellites with its GPS receiver. The tracker then uses its built-in radio transmitter to transmit its position to a network. The network can be either ground or satellite-based. Examples of ground-based networks include cellphone towers and amateur radio APRS stations. Examples of satellite-based networks include Globalstar and Iridium satellites relaying your tracker's signal down to a gateway on earth. Both types of networks are connected to the world wide web. As long as you have access to the internet (by computer, smartphone, etc.) and everything is working as it should, you will be able to track your weather balloon payload. The weakest link in any tracking system is always between the tracker and the network. If a cellphone tracker lands in a rural area without network coverage, you'll lose your payload. If a satellite tracker's antenna is not pointed at the sky, the satellites in orbit will never receive the tracker's transmitted signal and you'll lose your payload.

Mobile Phone / GSM Tracker – NEVER under any circumstance use a tracker such as a smartphone, GSM tracker (Global System for Mobile Communications), Sim card enabled device, etc. to track a weather balloon. A cellphone or GSM tracker is any device that transmits its location over a cellular network. This includes smart phones with tracking apps as well as popular vehicle or pet GSM trackers that work by sending a message (a simple text message/SMS) over a cellular network. We only cover these types of trackers in this tutorial because they are sometimes used by uninformed hobbyists launching weather balloons. There are three main reasons why you should never use these types of trackers. First, per [FCC 22.925](#), it is illegal and will result in your receiving a fine from the FCC (Federal Communications Commission). Second, there's a good chance your payload will not be able to transmit its landing coordinates. Most payloads land in rural areas with limited or no cellular network coverage. Third, these types of trackers are not designed for extreme environments. The batteries on most smart phones and GSM trackers won't work at -60 C. Your tracker will shut down before your payload even lands. As with most trackers, the GPS receiver in these types of trackers are not specialized for high altitude flight and the tracker will stop updating when flying above roughly 18,000 m.



three options have been successfully tested. Good deals on these trackers can also be found on eBay, Facebook Marketplace, etc. Make sure you test your tracker before flying, even when purchased new.

Satellite Tracker – This is by far the best option for tracking your weather balloon payload. Satellite trackers are designed to be both rugged and reliable. Unlike cellphone and APRS trackers, satellite trackers rely on a network of satellites in orbit to receive their position signal. These satellites then relay the signal to a gateway on earth which is connected to the internet. This allows your tracker to work just about anywhere on our planet. As long as your tracker has a clear view of the sky, a satellite in orbit will receive your tracker's position signal. There are a few things to keep in mind when using a satellite tracker. A satellite tracker's antenna must always be pointed at sky. Many people use Styrofoam coolers as their payload enclosures which have a tendency to roll onto their sides, or even upside down, when they land. The trackers antenna is no longer pointed at the sky and the satellites in orbit never receive your tracker's position signal. Many payloads are lost this way. Satellite trackers also require a subscription fee. Just as you pay your cellular network provider a monthly subscription fee, you also have to pay your satellite tracker service provider a monthly or annual subscription fee. Satellite trackers typically update their position once every 5 or 10 minutes. Satellite trackers also do not use specialized GPS receivers (as discussed above) and therefore typically stop updating above 18,000 m. Once the balloon bursts and the payload descends below the GPS lockout altitude, tracking will resume. Even though satellite trackers have a few limitations, (antenna must be pointed at the sky, low update rate, no specialized GPS receiver) no payload should fly without one. If you want to maximize your chance of recovering your payload, use a satellite tracker and use it correctly (no Styrofoam cooler payloads). The Spot Trace Satellite Tracker is one of the most popular tracking solutions for high altitude balloon flights. This exceptional piece of hardware communicates directly with satellites in orbit allowing you to track your payload just about anywhere on (or off) the planet. An annual service subscription is required. Visit www.findmespot.com for details. Due to supply chain issues, the Spot Trace may not be available. We also recommend considering the Spot II Satellite Tracker, or Delorme inReach Tracker. All



APRS Tracker – Although APRS trackers are not as reliable as a satellite trackers when it comes to payload recovery, it remains our favorite type of tracker. APRS trackers transmit their location to a network of repeaters and internet gates managed by amateur radio operators across North America and other parts of the world. To legally use an APRS tracker, the FCC does require that you have an amateur radio license. This is a straightforward process. Here's a [great article](#) on obtaining your license written by an elementary school educator. Most APRS trackers are designed for tracking vehicles. Their GPS receivers therefore have the same issue of not working above 18,000 m as satellite trackers do. Stratogear, however, has developed an APRS tracker specifically for high altitude ballooning called the **StratoTrack** (see image). Because the StratoTrack uses an advanced GPS receiver, its APRS transmitter is able to transmit your payload's position at any altitude, even above 18,000 m. Unlike satellite trackers, there is no subscription fee. The StratoTrack update rate is once per minute instead of once every 5 or 10 minutes. Most satellite trackers transmit only position, not altitude. You know where your payload is but you don't know how high it is. The StratoTrack transmits both position and altitude. It even transmits speed, battery voltage, temperature, and pressure data. This allows you to determine if your payload is ascending, close to burst altitude, descending, close to landing, etc. It also allows you to perform advanced analysis of the upper atmosphere. All this data can be viewed real-time on mystratotrack.com. You can even download your flight data into an excel file for additional analysis using mystratotrack.com. There is, however, a drawback to using an APRS transmitter. If your payload lands in a rural area far from an amateur radio station that can receive your tracker's signal, you will never know where your payload has landed. This is why you should never rely on your APRS tracker for locating your payload once it has landed. You should always use a satellite tracker that isn't limited by a ground network. View your APRS tracker as a backup which supplements your satellite tracker by providing additional information during your payload's flight (higher update rate including altitude, wind speed, temperature, pressure, etc.).

Next

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